

Intro Real Analysis

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April 2022

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1 Notions from Set Theory

1.1 Sets and Subsets

A set is determined entirely by its elements. We write $x \in S$ when x is an element of S , and $x \notin S$ otherwise. Thus

$$X = Y \iff \forall x (x \in X \Leftrightarrow x \in Y).$$

The singleton $\{x\}$ is the set whose only element is x ; it is not the same object as x . The empty set is denoted by $\emptyset$.

Sets are commonly given by listing elements or by a defining property:

$$\{x_1, x_2, x_3\}, \quad \{x \in S : x \text{ satisfies a stated property}\}.$$

For example, $\{x \in \mathbb{R} : 0 \leq x \leq 1\}$ is the closed unit interval.

Definition 1.1 (Subset). For sets X and S ,

$$X \subseteq S \iff \forall x (x \in X \Rightarrow x \in S).$$

If $X \subseteq S$ and $X \neq S$, then X is a *proper subset* of S , written $X \subsetneq S$.

Both $S \subseteq S$ and $\emptyset \subseteq S$ always hold. Keep membership and containment separate: $2 \in \{1, 2, 3\}$, but $\{2, 3\} \subseteq \{1, 2, 3\}$.

1.2 Operations on Sets

Definition 1.2 (Basic operations). For sets X, Y in an ambient set S , define

$$\begin{aligned} X \cap Y &= \{x : x \in X \text{ and } x \in Y\}, & X \cup Y &= \{x : x \in X \text{ or } x \in Y\}, \\ S \setminus X &= \{x \in S : x \notin X\}, & X \setminus Y &= \{x \in X : x \notin Y\}. \end{aligned}$$

Here “or” is inclusive. When S is understood, $S \setminus X$ is also written X^c .

Two useful identities, valid for $X, Y \subseteq S$, are De Morgan’s laws:

$$(X \cup Y)^c = X^c \cap Y^c, \quad (X \cap Y)^c = X^c \cup Y^c.$$

Sets are *disjoint* if their intersection is empty.

For an indexed family $\{X_i\}_{i \in I}$,

$$\bigcap_{i \in I} X_i = \{x : x \in X_i \text{ for every } i \in I\}, \quad \bigcup_{i \in I} X_i = \{x : x \in X_i \text{ for some } i \in I\}.$$

Consequently, if all $X_i \subseteq S$, then

$$\left(\bigcap_{i \in I} X_i \right)^c = \bigcup_{i \in I} X_i^c.$$

Definition 1.3 (Ordered pairs and products). An ordered pair satisfies $(a, b) = (c, d)$ exactly when $a = c$ and $b = d$. One set-theoretic model is

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

The Cartesian product of X and Y is

$$X \times Y = \{(x, y) : x \in X, y \in Y\}.$$

1.3 Functions

Definition 1.4 (Function). A function $f : X \rightarrow Y$ assigns to each $x \in X$ a unique element $f(x) \in Y$. The set X is the domain, Y is the codomain, and two functions $f, g : X \rightarrow Y$ are equal precisely when $f(x) = g(x)$ for all $x \in X$.

The graph of $f : X \rightarrow Y$ is

$$\Gamma_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y.$$

Conversely, a subset $\Gamma \subseteq X \times Y$ is the graph of a function $X \rightarrow Y$ iff for each $x \in X$ there is exactly one $y \in Y$ with $(x, y) \in \Gamma$.

Restrictions, enlargements of codomain, and compositions are written

$$f|_{X'} : X' \rightarrow Y, \quad g \circ f : X \rightarrow Z, \quad (g \circ f)(x) = g(f(x)).$$

The identity on X is $i_X(x) = x$.

Definition 1.5 (Injective, surjective, bijective). For $f : X \rightarrow Y$:

- (i) f is *injective* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$;
- (ii) f is *surjective* if for every $y \in Y$ there exists $x \in X$ with $f(x) = y$;
- (iii) f is *bijective* if it is both injective and surjective.

A bijection has an inverse $f^{-1} : Y \rightarrow X$ satisfying $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$. For arbitrary $f : X \rightarrow Y$ and subsets $A \subseteq X$, $B \subseteq Y$,

$$f(A) = \{f(x) : x \in A\}, \quad f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

The second expression is the *preimage* of B and does not require f to be invertible.

1.4 Finite and Infinite Sets

Let $\mathbb{N} = \{1, 2, 3, \dots\}$.

Definition 1.6 (Finite set). A set X is *finite* if $X = \emptyset$ or if X is in bijection with $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. In the latter case $|X| = n$.

A set is *infinite* if it is not finite. Proper subsets of finite sets have strictly fewer elements.

Theorem 1.1. *A set is infinite if and only if it is in bijection with a proper subset of itself.*

Proof. A finite set cannot be bijective with a proper subset, since proper subsets have fewer elements.

Conversely, let X be infinite. Choose distinct elements $x_1, x_2, x_3, \dots$ in X , and put

$$Y = X \setminus \{x_1, x_2, x_3, \dots\}.$$

Define

$$f : X \rightarrow \{x_2, x_3, x_4, \dots\} \cup Y, \quad f(x_k) = x_{k+1}, \quad f(y) = y \ (y \in Y).$$

This shifts the chosen sequence forward and fixes every element of Y , hence is a bijection from X onto the proper subset $\{x_2, x_3, \dots\} \cup Y$. $\square$

An n -tuple in X is a function $\{1, \dots, n\} \rightarrow X$; a sequence in X is a function $\mathbb{N} \rightarrow X$, usually written $(x_1, x_2, x_3, \dots)$.

2 The Real Number System

2.1 The Field Properties

Analysis rests on the exact algebraic and order structure of the real numbers. We begin by recording the axioms they satisfy.

Definition 2.1 (Group). A *group* is a nonempty set G equipped with a binary operation $\cdot$ such that:

1. $x \cdot y \in G$ for all $x, y \in G$ (closure).
2. $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in G$ (associativity).
3. There exists $e \in G$ with $e \cdot x = x \cdot e = x$ for every $x \in G$ (identity).
4. Each $x \in G$ has $x^{-1} \in G$ satisfying $x \cdot x^{-1} = x^{-1} \cdot x = e$ (inverse).

The group is *Abelian* if additionally $x \cdot y = y \cdot x$ for all x, y .

Definition 2.2 (Field). A *field* is a set $\mathbb{F}$ together with two binary operations, addition and multiplication, such that

1. $(\mathbb{F}, +)$ is an Abelian group with identity $0_{\mathbb{F}}$.
2. $(\mathbb{F} \setminus \{0_{\mathbb{F}}\}, \cdot)$ is an Abelian group with identity $1_{\mathbb{F}}$.
3. Multiplication distributes over addition:

$$x \cdot (y + z) = x \cdot y + x \cdot z, \quad (x + y) \cdot z = x \cdot z + y \cdot z.$$

Definition 2.3 (Real Numbers). The *real number system* $\mathbb{R}$ is the unique (up to isomorphism) complete ordered field. Practically we view $\mathbb{R}$ as a set endowed with addition, multiplication, and an order relation satisfying the axioms listed in this section.

Besides $\mathbb{R}$, the rationals $\mathbb{Q}$ and complexes $\mathbb{C}$ are important fields. Finite examples also exist (for instance $\mathbb{Z}/p\mathbb{Z}$ when p is prime), but they lack the order and completeness required for analysis.

Field axioms. Rosenlicht enumerates seven explicit properties; we collect them in two groups.

- (F1) **Commutativity.** $a + b = b + a$ and $ab = ba$.
- (F2) **Associativity.** $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
- (F3) **Identities.** There exist 0 and 1 with $a + 0 = a$ and $a \cdot 1 = a$.
- (F4) **Inverses.** Every a has $-a$ with $a + (-a) = 0$; each $a \neq 0$ has a^{-1} with $aa^{-1} = 1$.
- (F5) **Distributivity.** $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

These axioms imply all of the familiar algebraic rules. We spell out the basic consequences because they are used constantly.

Lemma 2.1 (Cancellation). *If $(G, \cdot)$ is a group and $xy = xz$, then $y = z$; likewise $yx = zx$ implies $y = z$.*

Proof. Multiply $xy = xz$ on the left by x^{-1} to obtain $y = z$. □

Corollary 2.0.1. *The identity element of a group is unique, inverses are unique, and $(x^{-1})^{-1} = x$.*

Proposition 2.1 (Algebraic identities). *Let $(\mathbb{F}, +, \cdot)$ be a field. For any $x, y \in \mathbb{F}$ the following hold.*

1. $0 \cdot x = x \cdot 0 = 0$ and $(-1)x = x(-1) = -x$.
2. $xy = 0$ implies $x = 0$ or $y = 0$.
3. $(-x)(-y) = xy$ and $-(x + y) = (-x) + (-y)$.
4. If $x, y \neq 0$ then $(xy)^{-1} = y^{-1}x^{-1}$.

Proof. Each statement is a short application of the axioms and Lemma 2.1. For instance,

$$x \cdot 0 = x \cdot (0 + 0) = x \cdot 0 + x \cdot 0,$$

so subtracting $x \cdot 0$ from both sides shows $x \cdot 0 = 0$. The other items are similar. $\square$

2.2 Order

To do analysis, the field must also be ordered.

Definition 2.4 (Ordered field). An *ordered field* is a field endowed with a relation $<$ such that for all x, y, z :

1. Exactly one of $x < y$, $x = y$, or $x > y$ holds (*trichotomy*).
2. If $x < y$, then $x + z < y + z$.
3. If $0 < z$ and $x < y$, then $xz < yz$.

We write $x \leq y$ when $x < y$ or $x = y$ and denote the set of positive elements by $\mathbb{F}_{>0} = \{x : x > 0\}$. The order interacts with addition and multiplication exactly as intuition dictates.

Lemma 2.2 (Order manipulations). *If $x > 0$ and $y > 0$, then $x + y > 0$ and $xy > 0$. If $x < y$ and $z < w$, then $x + z < y + w$. Multiplying by a negative element reverses inequalities.*

In an ordered field we define intervals, rays, and the absolute value $|x| = \max\{x, -x\}$. These capture the geometric idea of distance on the number line.

2.3 Completeness and Bounds

The property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$ is completeness.

Definition 2.5 (Bounds). Let $S \subseteq \mathbb{R}$. A number u is an *upper bound* if $x \leq u$ for all $x \in S$. If u is an upper bound and $u \leq v$ for every upper bound v , then u is the *supremum* $\sup S$. Lower bounds and the infimum $\inf S$ are defined similarly.

Theorem 2.1 (Least Upper Bound Axiom). *Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$; every nonempty subset bounded below has an infimum.*

Theorem 2.2 (Monotone convergence). *If $\{a_n\}$ is increasing and bounded above, then $a_n \rightarrow \sup\{a_n\}$.*

Proof. Let $s = \sup\{a_n\}$. Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so some N satisfies $a_N > s - \varepsilon$. For $n \geq N$ we have $s - \varepsilon < a_n \leq s$, proving $a_n \rightarrow s$. $\square$

Theorem 2.3 (Archimedean property). *For every $x > 0$ there exists $n \in \mathbb{N}$ with $n > x$.*

Proof. Suppose no such n exists. Then the set $\mathcal{A} = \{n : n \in \mathbb{N}\}$ is bounded above by x , so $s = \sup \mathcal{A}$ exists. But $s - 1$ is not an upper bound, so some $m \in \mathbb{N}$ satisfies $m > s - 1$, i.e., $m + 1 > s$. Since $m + 1 \in \mathbb{N}$, we contradict the definition of s . $\square$

Theorem 2.4 (Density of $\mathbb{Q}$). *If $a < b$ in $\mathbb{R}$, then there exists $q \in \mathbb{Q}$ with $a < q < b$.*

Proof. Consider $(b - a) > 0$. By the Archimedean property choose n with $n(b - a) > 1$. There exists an integer m such that $na < m < nb$, and then $m/n \in \mathbb{Q}$ satisfies $a < m/n < b$. $\square$

2.4 The Existence of Square Roots

Completeness quickly yields the fundamental fact that positive numbers possess square roots.

Theorem 2.5. *For every $a > 0$ there exists a unique $b > 0$ such that $b^2 = a$.*

Proof. Consider the set $S = \{x \in \mathbb{R} : x \geq 0 \text{ and } x^2 \leq a\}$. It is nonempty (since $0 \in S$) and bounded above (for example by $a + 1$). Let $b = \sup S$. We first show $b^2 \leq a$. Otherwise $b^2 > a$ and putting $\varepsilon = b^2 - a > 0$ we may take $\eta = \varepsilon/(2b)$; note that $b > 0$ because $a > 0$. Then $(b - \eta)^2 = b^2 - 2b\eta + \eta^2 = b^2 - \varepsilon + \eta^2 > a$, so every $x \in S$ satisfies $x < b - \eta$. That means $b - \eta$ is an upper bound smaller than b , contradicting the definition of b . Hence $b^2 \leq a$.

If $b^2 < a$, set $\varepsilon = a - b^2 > 0$ and choose $\eta = \min\{1, \varepsilon/(2b + 1)\}$. Then

$$(b + \eta)^2 = b^2 + 2b\eta + \eta^2 \leq b^2 + \eta(2b + 1) \leq b^2 + \varepsilon = a,$$

so $b + \eta \in S$. This contradicts b being an upper bound. Therefore $b^2 = a$.

Uniqueness follows from the order axioms: if $c > 0$ also satisfies $c^2 = a$, then $(b - c)(b + c) = 0$. Proposition 2.1 implies either $b = c$ or $b = -c$. The latter is impossible because $b, c > 0$, so $b = c$. $\square$

The existence of square roots supplies the positive branch of the absolute value and, together with the field axioms, completes the algebraic description of $\mathbb{R}$.

3 Metric Spaces

3.1 Definition and Examples

Metric spaces generalize familiar geometric notions of distance.

Definition 3.1 (Metric Space). A *metric space* is a pair (M, d) consisting of a set M and a function $d: M \times M \rightarrow \mathbb{R}$ satisfying for all $x, y, z \in M$:

1. $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$.
2. $d(x, y) = d(y, x)$.
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

Lemma 3.1. *Nonnegativity is automatic: $d(x, y) = \frac{1}{2}(d(x, y) + d(y, x)) \geq \frac{1}{2}d(x, x) = 0$.*

Theorem 3.1 (General Triangle Inequality). *Given points $x_1, \dots, x_n$ in (M, d) ,*

$$d(x_1, x_n) \leq \sum_{k=1}^{n-1} d(x_k, x_{k+1}).$$

Corollary 3.1.1 (Reverse Triangle Inequality). *For all $x, y, z \in M$, $|d(x, z) - d(z, y)| \leq d(x, y)$.*

These inequalities turn the metric into a powerful bookkeeping device for estimates.

Examples.

1. On $\mathbb{R}^n$ the Euclidean metric $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ arises from the dot product. The Cauchy-Schwarz inequality

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

implies $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$, which provides the triangle inequality for d .

2. The *taxicab* (or ℓ^1) metric on $\mathbb{R}^n$ is $d_1(\mathbf{x}, \mathbf{y}) = \sum |x_i - y_i|$; each coordinate satisfies the one-dimensional triangle inequality, so the sum does as well.
3. The *discrete* metric on an arbitrary set M is $d(x, y) = 0$ when $x = y$ and $d(x, y) = 1$ otherwise.

Subspaces. If $E \subseteq M$, the restriction of d to $E \times E$ makes E into a metric space; this is the *subspace metric*.

3.2 Open Sets, Closed Sets, and Neighborhoods

For $p \in M$ and $r > 0$ define the open ball $B_r(p) = \{x : d(x, p) < r\}$ and the closed ball $\overline{B}_r(p) = \{x : d(x, p) \leq r\}$.

Definition 3.2. A subset $U \subseteq M$ is *open* if for every $x \in U$ there exists $r > 0$ with $B_r(x) \subseteq U$. A subset $F \subseteq M$ is *closed* if its complement is open. The empty set and M are both open and closed.

Open sets collect points together with a “cushion” around them. Closed sets are precisely those that contain all limit points of sequences within them (see Proposition 3.2).

Proposition 3.1. *Open balls are open, closed balls are closed, and finite intersections of open (resp. closed) balls are open (resp. closed).*

Proof. If $x \in B_r(p)$ set $\rho = r - d(x, p) > 0$. Whenever $y \in B_\rho(x)$, the triangle inequality gives $d(y, p) \leq d(y, x) + d(x, p) < \rho + d(x, p) = r$, so $y \in B_r(p)$. The closed case follows by considering complements. $\square$

Theorem 3.2. *Let $\{U_\alpha\}_{\alpha \in A}$ be open subsets of M and $F_1, \dots, F_n$ closed subsets.*

1. $\bigcup_{\alpha \in A} U_\alpha$ is open.
2. $\bigcap_{k=1}^n U_k$ is open for any finite subcollection.
3. $\bigcap_{\alpha \in A} F_\alpha$ is closed and $\bigcup_{k=1}^n F_k$ is closed.

Limit points, interior, and closure.

Definition 3.3. A point x is a *limit point* of $E \subseteq M$ if every ball about x intersects $E \setminus \{x\}$. The set of all limit points is denoted E' . The *closure* $\overline{E} = E \cup E'$ is the smallest closed set containing E , and the *interior* $\text{int}(E)$ is the largest open set contained in E .

Proposition 3.2. *E is closed if and only if it contains its limit points; equivalently $E = \overline{E}$.*

Proof. If E is closed and $x \in E'$ then every ball around x meets E , hence x cannot lie in the open complement. Conversely, if E contains its limit points, E^c is open: given $x \in E^c$, there exists $r > 0$ with $B_r(x) \cap E = \emptyset$, so $B_r(x) \subseteq E^c$. $\square$

Proposition 3.3. *In $\mathbb{R}$ the open balls are intervals (a, b) , while closed balls are $[a, b]$. In $\mathbb{R}^n$ rectangular boxes $\prod_{i=1}^n (a_i, b_i)$ are open, their closures $\prod [a_i, b_i]$ are closed, and both are bounded.*

3.3 Boundedness and Extremal Points

Definition 3.4 (Bounded Set). A subset S is *bounded* if $S \subseteq \overline{B}_r(p)$ for some p and r .

Balls witness boundedness, so finite unions of bounded sets are bounded. In $\mathbb{R}^n$ closed rectangles are bounded because each coordinate difference is bounded.

Theorem 3.3. *A nonempty closed subset of $\mathbb{R}$ that is bounded above has a maximum; one bounded below has a minimum.*

3.4 Convergent Sequences

Definition 3.5 (Convergence). A sequence $\{p_n\}$ converges to p if for every $\varepsilon > 0$ there exists N with $d(p_n, p) < \varepsilon$ for $n \geq N$.

Limits are unique. A sequence converges to p if and only if every subsequence has a further subsequence converging to p . Open and closed sets can be characterized via sequences:

Proposition 3.4. 1. U is open if and only if no sequence entirely outside U converges to a point of U .
2. F is closed if and only if whenever $p_n \in F$ and $p_n \rightarrow p$, then $p \in F$.

3.5 Completeness

Definition 3.6 (Cauchy Sequence). A sequence $\{p_n\}$ is *Cauchy* if for every $\varepsilon > 0$ there exists N with $d(p_n, p_m) < \varepsilon$ for all $m, n \geq N$. A metric space is *complete* if every Cauchy sequence converges.

Proposition 3.5. $\mathbb{R}^n$ with the Euclidean metric is complete.

Proof. Each coordinate of a Cauchy sequence is Cauchy in $\mathbb{R}$, hence convergent. The coordinatewise limit lies in $\mathbb{R}^n$ and is the limit of the vector sequence. $\square$

Theorem 3.4 (Completion criterion). *A subspace $E \subseteq M$ is complete if and only if it is closed in M (when M itself is complete).*

3.6 Compactness

Definition 3.7 (Compactness). A subset $K \subseteq M$ is *compact* if every open cover $\{U_\alpha\}$ of K admits a finite subcover. Equivalently, K is compact if every sequence in K has a convergent subsequence with limit in K (sequential compactness).

Theorem 3.5 (Heine–Borel). *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proposition 3.6. *Closed subsets of compact sets are compact, and continuous images of compact sets are compact.*

3.7 Connectedness

Definition 3.8 (Connected Set). A metric space M is *connected* if it cannot be written as the union of two disjoint nonempty open sets. A subset $E \subseteq M$ is connected if it is connected with the subspace metric.

Proposition 3.7. *Intervals in $\mathbb{R}$ are connected, and any connected subset of $\mathbb{R}$ is an interval. Products of connected sets are connected. Continuous images of connected sets are connected.*

Connectedness ensures that intermediate values cannot be skipped; this feeds directly into the Intermediate Value Theorem in the next chapter.

4 Continuous Functions

Continuous functions will carry the weight of the topological results developed in Chapter 3. Rosenlicht treats functions between arbitrary metric spaces; the real- and vector-valued cases fall out as specializations.

4.1 Definition of Continuity. Examples

Let (E, d) and (E', d') be metric spaces.

Definition 4.1 (Continuity at a point). A function $f: E \rightarrow E'$ is *continuous at* $p_0 \in E$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $d(p, p_0) < \delta$ implies $d'(f(p), f(p_0)) < \varepsilon$. It is *continuous* if it is continuous at every point.

The definition admits two equivalent reformulations.

Proposition 4.1 (Neighborhood formulation). *The following are equivalent for $f: E \rightarrow E'$ and $p_0 \in E$.*

1. f is continuous at p_0 .
2. For every open set $U \subseteq E'$ with $f(p_0) \in U$, there exists an open set $V \subseteq E$ containing p_0 such that $f(V) \subseteq U$.
3. For every sequence $p_n \rightarrow p_0$ in E , we have $f(p_n) \rightarrow f(p_0)$ in E' .

Corollary 4.0.1. *f is continuous if and only if the preimage of every open set is open (equivalently, the preimage of every closed set is closed).*

These criteria prove continuity for the functions encountered in calculus: polynomials are continuous everywhere, rational functions are continuous on their domain, absolute value and maximum/minimum are continuous combinations of basic operations, and so on.

4.2 Continuity and Limits

Continuity aligns perfectly with function limits.

Theorem 4.1 (Sequential characterization). *If $f: E \rightarrow E'$ and $p \in E$, then f is continuous at p if and only if for every sequence $p_n \rightarrow p$ we have $f(p_n) \rightarrow f(p)$.*

Theorem 4.2 (Composition and restriction). *If $f: E \rightarrow E'$ is continuous at $p \in E$ and $g: E' \rightarrow E''$ is continuous at $f(p)$, then $g \circ f$ is continuous at p . If $A \subseteq E$ carries the subspace metric and f is continuous on E , then the restriction $f|_A$ is continuous on A .*

Proposition 4.2 (Gluing lemma). *If $E = F \cup G$ with F, G closed subsets and $f: E \rightarrow E'$ agrees with continuous maps on F and on G , then f is continuous.*

Continuity also respects monotone bijections between intervals: if $U, V \subseteq \mathbb{R}$ are intervals and $f: U \rightarrow V$ is strictly increasing and onto, then both f and f^{-1} are continuous.

4.3 Rational Operations and Vector-Valued Functions

The algebra of continuous real-valued functions mimics the algebra of real numbers.

Theorem 4.3 (Algebra of continuous functions). *Let $f, g: E \rightarrow \mathbb{R}$ be continuous and let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Then the functions $f + g$, $f - g$, fg , and $\varphi \circ f$ are continuous. If $g(p) \neq 0$ for all p , then f/g is continuous.*

Corollary 4.3.1. *Polynomials and rational functions (on the domain where the denominator is nonzero) are continuous. If $\mathbf{f}: E \rightarrow \mathbb{R}^n$ is given by $\mathbf{f}(p) = (f_1(p), \dots, f_n(p))$, then $\mathbf{f}$ is continuous if and only if each coordinate function f_k is continuous.*

4.4 Continuous Functions on Compact Metric Spaces

Compactness translates topological information into quantitative statements.

Theorem 4.4 (Continuous images of compact sets). *If $K \subseteq E$ is compact and $f: E \rightarrow E'$ is continuous, then $f(K) \subseteq E'$ is compact.*

Specializing to real-valued functions produces the classical extreme value and boundedness results.

Corollary 4.4.1. *If K is compact and $f: K \rightarrow \mathbb{R}$ is continuous, then f is bounded and attains both its maximum and minimum on K .*

Theorem 4.5 (Heine–Cantor). *Continuous functions on compact metric spaces are uniformly continuous.*

Proof. Suppose f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ and points $p_n, q_n \in K$ with $d(p_n, q_n) \rightarrow 0$ but $|f(p_n) - f(q_n)| \geq \varepsilon_0$. Compactness provides a convergent subsequence $p_{n_k} \rightarrow p$; the paired points q_{n_k} also converge to p . Continuity contradicts the fixed gap ε_0 . $\square$

Uniform continuity guarantees that Cauchy sequences of function values behave well. In particular, the sup metric

$$d_\infty(f, g) = \sup_{p \in K} d'(f(p), g(p))$$

turns $C(K, E')$ (continuous maps $K \rightarrow E'$) into a metric space. When K is compact and E' is complete, $C(K, E')$ is complete under d_∞ : a uniformly Cauchy sequence of continuous functions converges uniformly to a continuous function.

4.5 Continuous Functions on Connected Metric Spaces

Connectedness rules out “jumps”.

Theorem 4.6 (Image of a connected set). *If E is connected and $f: E \rightarrow E'$ is continuous, then $f(E)$ is connected.*

In $\mathbb{R}$ the connected subsets are intervals; applying Theorem 4.6 yields the Intermediate Value Theorem.

Corollary 4.6.1 (Intermediate Value Theorem). *If $f: [a, b] \rightarrow \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ with $f(c) = y$.*

Further consequences include: a continuous function on an interval that never changes sign is either strictly positive, strictly negative, or identically zero; a continuous bijection from a compact interval to an interval is monotone with a continuous inverse.

4.6 Sequences of Functions

Finally we examine sequences of continuous functions. Let (f_n) be functions $E \rightarrow E'$.

Definition 4.2 (Uniform convergence). f_n converges *uniformly* to f if for every $\varepsilon > 0$ there exists N with $d'(f_n(p), f(p)) < \varepsilon$ for all $p \in E$ whenever $n \geq N$.

Theorem 4.7 (Uniform limits preserve continuity). *If each f_n is continuous and $f_n \rightarrow f$ uniformly on E , then f is continuous.*

Proof. Given $\varepsilon > 0$, choose N such that $d'(f_n(p), f(p)) < \varepsilon/3$ for all p and $n \geq N$. Continuity of f_N provides $\delta > 0$ so that $d(p, q) < \delta$ implies $d'(f_N(p), f_N(q)) < \varepsilon/3$. For $d(p, q) < \delta$,

$$d'(f(p), f(q)) \leq d'(f(p), f_N(p)) + d'(f_N(p), f_N(q)) + d'(f_N(q), f(q)) < \varepsilon,$$

so f is continuous. □

Uniform convergence is naturally encoded by the sup metric on $C(K, E')$ discussed earlier: $f_n \rightarrow f$ uniformly if and only if $d_\infty(f_n, f) \rightarrow 0$. When K is compact and E' complete, $C(K, E')$ is itself complete under d_∞ .

Examples include Fourier-type partial sums converging uniformly on compact subsets, or piecewise linear approximations to continuous paths. This perspective will be revisited in the later chapters on series and approximations.

5 Differentiation

Differentiation returns us to single-variable calculus, but armed with the metric machinery from the previous chapters we can present the proofs succinctly.

5.1 Definition of the Derivative

Let $U \subseteq \mathbb{R}$ be open and $f: U \rightarrow \mathbb{R}$.

Definition 5.1 (Derivative). f is *differentiable* at $x_0 \in U$ if the limit

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. The function f' defined on the set of points where the limit exists is the *derivative* of f .

Equivalently, differentiability means there exists a linear function $L(x) = f(x_0) + f'(x_0)(x - x_0)$ such that

$$\lim_{x \rightarrow x_0} \frac{f(x) - L(x)}{x - x_0} = 0.$$

The graph of f near x_0 therefore hugs its tangent line.

Proposition 5.1. *If f is differentiable at x_0 , then f is continuous at x_0 .*

Proof. Using the definition with ε - δ notation,

$$|f(x) - f(x_0)| \leq |x - x_0| \left| \frac{f(x) - f(x_0)}{x - x_0} \right|,$$

and the quotient tends to $|f'(x_0)|$, so $f(x) \rightarrow f(x_0)$. □

5.2 Rules of Differentiation

The derivative behaves linearly and obeys the product and chain rules familiar from calculus.

Theorem 5.1 (Algebraic rules). *If f, g are differentiable at x_0 and $c \in \mathbb{R}$, then so are $f + g$, cf , and fg , with*

$$(f + g)'(x_0) = f'(x_0) + g'(x_0), \quad (cf)'(x_0) = cf'(x_0), \quad (fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0).$$

If $g(x_0) \neq 0$, then $\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$.

Theorem 5.2 (Chain rule). *If f is differentiable at x_0 and g is differentiable at $f(x_0)$, then $g \circ f$ is differentiable at x_0 with*

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

Using these rules we recover the standard derivatives: polynomials differentiate via the binomial theorem, rational functions differentiate wherever they are defined, and so on.

5.3 The Mean Value Theorem

We now reach the central estimate for differentiable functions.

Theorem 5.3 (Rolle). *If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, differentiable on (a, b) , and satisfies $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Proof. $[a, b]$ is compact, so f attains a maximum and minimum. If either extremum occurs at an interior point c , then near c the function cannot increase or decrease, forcing $f'(c) = 0$. Otherwise f is constant. $\square$

Theorem 5.4 (Mean Value Theorem). *If $f: [a, b] \rightarrow \mathbb{R}$ is continuous and differentiable on (a, b) , then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Apply Rolle's theorem to $g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$ after noting that $g(a) = g(b)$. $\square$

Corollary 5.4.1. *If $f'(x) = 0$ for all x in an interval I , then f is constant on I .*

Corollary 5.4.2. *If $f'(x) \geq 0$ (respectively > 0) on an interval, then f is nondecreasing (respectively strictly increasing).*

Corollary 5.4.3 (Cauchy mean value theorem). *If f, g are continuous on $[a, b]$ and differentiable on (a, b) with $g'(x) \neq 0$, then there exists c such that*

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

The mean value theorem underlies standard inequalities, e.g., the Lipschitz bound $|f(b) - f(a)| \leq M|b - a|$ when $|f'| \leq M$ on the interval, and L'Hospital's rule for $0/0$ forms (derived using Cauchy's version).

5.4 Taylor's Theorem

Repeated differentiation grants finer control via Taylor polynomials.

Definition 5.2 (Higher derivatives). If $f^{(n-1)}$ exists on an open interval, we say f is n times differentiable when $f^{(n-1)}$ is differentiable; the new derivative is $f^{(n)}$.

Theorem 5.5 (Taylor's formula with remainder). Suppose f has $n+1$ continuous derivatives on $[a, b]$. For any $x \in [a, b]$ there exists ξ between x and a such that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Define P_n to be the n th-degree Taylor polynomial at a . Consider

$$g(t) = f(t) - P_n(t) - \frac{f^{(n+1)}(x)}{(n+1)!}(t-a)^{n+1}.$$

Rolle's theorem applied repeatedly to appropriate combinations of g shows that $g^{(n+1)}$ vanishes somewhere between a and x , yielding the claimed ξ . $\square$

Corollary 5.5.1 (Estimates). If $|f^{(n+1)}(t)| \leq M$ on $[a, b]$, then

$$\left| f(x) - \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x-a)^k \right| \leq \frac{M}{(n+1)!} |x-a|^{n+1}.$$

Taylor's theorem recovers power series expansions for elementary functions and provides precise error estimates for approximations.

6 Riemann Integration

Riemann integration assigns areas to graphs of real-valued functions on intervals and underpins the rest of analysis.

6.1 Definition and Examples

Let $[a, b]$ be a closed interval.

Definition 6.1 (Partition). A *partition* of $[a, b]$ is a finite sequence $P = \{x_0, \dots, x_N\}$ with $a = x_0 < x_1 < \cdots < x_N = b$. Its *mesh* is $|P| = \max_{1 \leq i \leq N} (x_i - x_{i-1})$.

Definition 6.2 (Riemann sum). If $f : [a, b] \rightarrow \mathbb{R}$ and P is a partition, a *Riemann sum* of f with respect to P is

$$S(P, f) = \sum_{i=1}^N f(\xi_i)(x_i - x_{i-1}),$$

where each $\xi_i \in [x_{i-1}, x_i]$.

Definition 6.3 (Riemann integrability). f is *Riemann integrable* on $[a, b]$ if there exists $A \in \mathbb{R}$ such that for every $\varepsilon > 0$ there is $\delta > 0$ with $|S(P, f) - A| < \varepsilon$ whenever $|P| < \delta$. The number A is the Riemann integral $\int_a^b f(x) dx$.

Upper and lower Darboux sums provide a more practical criterion. For a partition P let

$$U(P, f) = \sum_{i=1}^N \left(\sup_{x \in [x_{i-1}, x_i]} f(x) \right) (x_i - x_{i-1}), \quad L(P, f) = \sum_{i=1}^N \left(\inf_{x \in [x_{i-1}, x_i]} f(x) \right) (x_i - x_{i-1}).$$

Then f is integrable if and only if $\sup_P L(P, f) = \inf_P U(P, f)$; the common value equals the integral. Continuous functions on $[a, b]$, monotone functions, and piecewise continuous functions are all integrable. Functions with jump discontinuities at finitely many points are integrable; the sign function is not integrable at a removable discontinuity unless we restrict the interval.

6.2 Linearity and Order Properties

Theorem 6.1 (Linearity). *If f, g are integrable on $[a, b]$ and $c \in \mathbb{R}$, then cf and $f + g$ are integrable, and*

$$\int_a^b cf(x) dx = c \int_a^b f(x) dx, \quad \int_a^b (f + g)(x) dx = \int_a^b f(x) dx + \int_a^b g(x) dx.$$

Theorem 6.2 (Order properties). *If $f \leq g$ on $[a, b]$, then $\int_a^b f \leq \int_a^b g$. Moreover*

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx \leq (b - a) \sup_{x \in [a, b]} |f(x)|.$$

Theorem 6.3 (Additivity). *If $a < c < b$ and f is integrable on $[a, b]$, then f is integrable on $[a, c]$ and $[c, b]$ and*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

6.3 Existence Criteria

To show f is integrable it suffices to show the upper and lower sums can be made arbitrarily close.

Proposition 6.1. *If f is bounded on $[a, b]$ and for every $\varepsilon > 0$ there exists $\delta > 0$ with $U(P, f) - L(P, f) < \varepsilon$ whenever $|P| < \delta$, then f is integrable.*

Continuity ensures this condition: given $\varepsilon > 0$, uniform continuity of f provides δ so that each subinterval of P with mesh $< \delta$ has oscillation $< \varepsilon/(b - a)$, forcing $U(P, f) - L(P, f) < \varepsilon$.

Monotone functions are integrable because the oscillation on each subinterval is at most the difference of endpoint values, which can be controlled by making the mesh small.

6.4 The Fundamental Theorem of Calculus

Integration and differentiation are inverse processes for continuous functions.

Theorem 6.4 (First Fundamental Theorem). *If f is integrable on $[a, b]$ and we define $F(x) = \int_a^x f(t) dt$, then F is continuous on $[a, b]$ and differentiable at every point where f is continuous; moreover $F'(x) = f(x)$ at such points.*

Proof. Continuity of F follows from the inequality $|F(x) - F(y)| \leq \int_y^x |f| \leq M|x - y|$ for $M = \sup |f|$. Differentiability uses the Mean Value Theorem and the fact that f is continuous at the point in question. $\square$

Theorem 6.5 (Second Fundamental Theorem). *If f is continuous on $[a, b]$ and F is any antiderivative of f , then*

$$\int_a^b f(x) dx = F(b) - F(a).$$

Proof. By Theorem 6.4, $F(x) = \int_a^x f(t) dt$ is an antiderivative. Any other antiderivative differs by a constant, so evaluating at $x = b$ gives the desired equality. $\square$

The Fundamental Theorem reduces many definite integrals to antiderivatives and shows that continuous functions possess antiderivatives.

6.5 Logarithms and Exponentials

As an application, we construct the logarithm and exponential functions via integrals.

Definition 6.4. For $x > 0$ define

$$\log x = \int_1^x \frac{1}{t} dt.$$

Then $\log$ is continuous, strictly increasing, and differentiable with $\log'(x) = 1/x$. Set $\exp$ to be the inverse function: $\exp(\log x) = x$. Extend $\exp$ to all real numbers via $\exp(x) = \sum_{n=0}^{\infty} x^n/n!$ if desired; it satisfies $\exp'(x) = \exp(x)$ and $\exp(\log x) = x$.

Proposition 6.2. *The function $\log$ satisfies $\log(xy) = \log x + \log y$, $\log(1) = 0$, and $\log(x^r) = r \log x$ for rational r . Exponentiation obeys $\exp(x + y) = \exp(x)\exp(y)$, $\exp(0) = 1$, and $\exp$ is the unique solution of $y' = y$, $y(0) = 1$.*

These definitions agree with the familiar power-series constructions and yield the usual properties rigorously from the Riemann integral.

7 Interchange of Limit Operations

Limit processes rarely occur in isolation. We often encounter sequences of functions for which we wish to integrate, differentiate, or otherwise take limits term by term. This chapter records the situations where interchanging these operations is justified.

7.1 Integration and Differentiation of Sequences of Functions

Let $\{f_n\}$ be functions on a closed interval $[a, b]$.

Theorem 7.1 (Integration of uniformly convergent sequences). *If each f_n is Riemann integrable on $[a, b]$ and $f_n \rightarrow f$ uniformly, then f is integrable and*

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx.$$

Proof. Uniform convergence implies the difference $|f_n - f|$ can be made uniformly small, hence the difference of integrals is bounded by $(b - a) \sup |f_n - f|$, which tends to zero. $\square$

Theorem 7.2 (Term-by-term differentiation). *Suppose each f_n is differentiable on $[a, b]$, that $(f_n(x_0))$ converges for some $x_0 \in [a, b]$, and that f'_n converges uniformly on $[a, b]$ to a function g . Then f_n converges uniformly to a differentiable function f , and $f' = g$.*

Proof. Integrating f'_n from x_0 to x shows $f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n(t) dt$. Uniform convergence of the derivatives implies uniform convergence of the integrals, hence the f_n converge uniformly to $f(x) = \lim f_n(x_0) + \int_{x_0}^x g(t) dt$, whose derivative is g . $\square$

7.2 Infinite Series

Series are sequences written additively. Let $\sum_{n=1}^{\infty} a_n$ denote the series with partial sums $s_n = \sum_{k=1}^n a_k$.

Definition 7.1. The series $\sum a_n$ *converges* if the sequence (s_n) converges. It converges *absolutely* if $\sum |a_n|$ converges.

Theorem 7.3 (Comparison tests). *If $|a_n| \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges absolutely. If $a_n \geq b_n \geq 0$ and $\sum b_n$ diverges, then so does $\sum a_n$.*

Theorem 7.4 (Alternating series test). *If (a_n) is decreasing to 0 and the terms alternate in sign, then $\sum (-1)^{n-1} a_n$ converges, and the truncation error is at most a_{n+1} in magnitude.*

Definition 7.2. A series of functions $\sum f_n$ converges *uniformly* on a set E if its sequence of partial sums converges uniformly. The Weierstrass M -test states that if $|f_n(x)| \leq M_n$ on E and $\sum M_n$ converges, then $\sum f_n$ converges uniformly.

Uniform convergence allows termwise integration and differentiation via Theorems 7.1 and 7.2.

7.3 Power Series

Power series provide a major class of function series.

Definition 7.3. A power series centered at a is $\sum_{n=0}^{\infty} c_n(x-a)^n$. The radius of convergence R is the unique extended real number such that the series converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$.

Theorem 7.5 (Cauchy–Hadamard formula). *The radius of convergence satisfies $1/R = \limsup_{n \rightarrow \infty} |c_n|^{1/n}$, with the conventions $R = \infty$ if the limsup is 0 and $R = 0$ if the limsup is ∞ .*

Theorem 7.6 (Termwise operations). *Inside the disk $|x-a| < R$ a power series may be differentiated and integrated term by term:*

$$\left(\sum_{n=0}^{\infty} c_n(x-a)^n \right)' = \sum_{n=1}^{\infty} n c_n(x-a)^{n-1}, \quad \int \sum_{n=0}^{\infty} c_n(x-a)^n dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}.$$

Both series share the same radius of convergence.

7.4 The Trigonometric Functions

The sine and cosine functions arise naturally from power series.

Definition 7.4. Define

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

These series converge for all $x \in \mathbb{R}$.

Proposition 7.1. *The functions $\sin$ and $\cos$ are entire (analytic everywhere) and satisfy*

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad \sin^2 x + \cos^2 x = 1.$$

Theorem 7.7 (Addition formulas). *For all $x, y \in \mathbb{R}$,*

$$\sin(x + y) = \sin x \cos y + \cos x \sin y, \quad \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

Proof. Expand the power series for $\sin(x + y)$ and $\cos(x + y)$ and rearrange terms; absolute convergence justifies the algebra. $\square$

7.5 Differentiation under the Integral Sign

Consider a function $F(t, x)$ defined on $[a, b] \times I$, where $I \subseteq \mathbb{R}$ is an interval, and define

$$\Phi(x) = \int_a^b F(t, x) dt.$$

Theorem 7.8 (Leibniz rule). *If $F(t, x)$ and $\partial F / \partial x$ are continuous on $[a, b] \times I$, then Φ is differentiable on I with*

$$\Phi'(x) = \int_a^b \frac{\partial F}{\partial x}(t, x) dt.$$

Proof. Apply the mean value theorem in the x -variable, uniformly in t , and use Theorem 7.1. $\square$

This formula is frequently quoted as “differentiating under the integral sign.” It allows parameter-dependent integrals to be handled by calculus on the parameter.

8 The Method of Successive Approximations

Iterative methods can solve a variety of problems—root finding, implicit equations, and differential equations. Rosenlicht distills the essential fixed-point principle and shows how it applies widely.

8.1 The Fixed Point Theorem

Theorem 8.1 (Banach fixed point theorem). *Let (E, d) be a metric space and $F: E \rightarrow E$ satisfy $d(F(p), F(q)) \leq \lambda d(p, q)$ for some $\lambda < 1$. Then F has a unique fixed point p^* (i.e., $F(p^*) = p^*$), and for any starting point p_0 the iterates $p_{n+1} = F(p_n)$ converge to p^* .*

Proof. The contraction condition ensures $d(p_{n+1}, p_n) \leq \lambda^n d(p_1, p_0)$, so (p_n) is Cauchy and hence convergent if E is complete. Passing to the limit in $p_{n+1} = F(p_n)$ yields $F(p^*) = p^*$. Uniqueness follows by applying the contraction inequality to two fixed points. $\square$

8.2 Simplest Case of the Implicit Function Theorem

Consider an equation $F(x, y) = 0$ with F continuous and continuously differentiable near (x_0, y_0) .

Theorem 8.2 (Implicit function, one variable). *If $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$, then there exists a neighborhood of x_0 on which the equation $F(x, y) = 0$ defines a unique differentiable function $y = \phi(x)$ with $\phi(x_0) = y_0$.*

Proof. Rewrite the equation as $y = y_0 - \frac{F(x, y)}{\partial F / \partial y(x_0, y_0)} + \text{error}$ and show this defines a contraction in y for x near x_0 . The fixed point of the resulting map is the desired $\phi(x)$. Differentiability follows by implicit differentiation. $\square$

8.3 Existence and Uniqueness for ODEs

The fixed-point method also yields solutions to first-order ordinary differential equations.

Theorem 8.3 (Picard–Lindelöf). *Let $f(x, y)$ be continuous in a rectangle $|x - x_0| \leq a$, $|y - y_0| \leq b$, and suppose f satisfies a Lipschitz condition in y : $|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2|$. Then the initial value problem $y' = f(x, y)$, $y(x_0) = y_0$ has a unique solution on some interval around x_0 .*

Proof. Integrate the differential equation to obtain the integral equation $y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$. Define an operator T on the space of continuous functions by $(Ty)(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$. Under the Lipschitz condition and for $|x - x_0|$ small enough, T is a contraction with respect to the sup metric, so it has a unique fixed point, which is the desired solution. $\square$

These iterative arguments justify the standard root-finding schemes (Newton’s method), guarantee the solvability of implicit equations, and provide existence and uniqueness for differential equations under natural hypotheses.

9 Partial Differentiation

Functions of several variables inherit many properties from one-variable calculus, but extending definitions carefully avoids pitfalls.

9.1 Definitions and Basic Properties

Let $U \subseteq \mathbb{R}^n$ be open and $f: U \rightarrow \mathbb{R}$.

Definition 9.1 (Partial derivatives). The i th partial derivative of f at $a \in U$ is

$$\frac{\partial f}{\partial x_i}(a) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(a)}{h},$$

if the limit exists.

Having all partial derivatives exist does not guarantee good behavior. The correct higher-dimensional analogue of differentiability is best phrased in terms of linear approximations.

Definition 9.2 (Differentiability). f is *differentiable* at a if there exists a linear map $L: \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\lim_{h \rightarrow 0} \frac{|f(a + h) - f(a) - L(h)|}{\|h\|} = 0.$$

This L is the differential df_a , and it is given by $L(h) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a)h_i$ when f is differentiable with existing partials.

Theorem 9.1. *If f is differentiable at a , then all partial derivatives exist at a and f is continuous at a .*

9.2 Higher Derivatives

Higher-order derivatives are defined by iterating the differential, provided the lower-order derivatives are differentiable.

Definition 9.3. If the first partial derivatives of f exist and are continuous on U , then f is said to be C^1 . Inductively, f is C^k if its partial derivatives up to order k exist and are continuous.

Mixed partial derivatives often coincide.

Theorem 9.2 (Equality of mixed partials). *If the second-order partial derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ are continuous near a , then they are equal at a .*

Higher-dimensional Taylor expansions mirror the one-variable case by replacing powers with multi-index notation.

Theorem 9.3 (Taylor's theorem in $\mathbb{R}^n$). *If f is C^{k+1} on a convex set containing a and $a + h$, then*

$$f(a + h) = f(a) + \sum_{|\alpha|=1}^k \frac{D^\alpha f(a)}{\alpha!} h^\alpha + R_{k+1}(h),$$

where $|R_{k+1}(h)| \leq C \|h\|^{k+1}$ for some constant C .

9.3 Implicit Function Theorem

The implicit function theorem generalizes the one-dimensional result.

Theorem 9.4 (Implicit function theorem). *Let $F: \mathbb{R}^{m+n} \rightarrow \mathbb{R}^n$ be C^1 near (x_0, y_0) with $F(x_0, y_0) = 0$. If the Jacobian matrix $\frac{\partial F}{\partial y}(x_0, y_0)$ is invertible, then there exist neighborhoods U of x_0 and V of y_0 and a unique C^1 function $\phi: U \rightarrow V$ such that $F(x, \phi(x)) = 0$ for all $x \in U$.*

Proof. Apply the contraction mapping principle to the map $y \mapsto y - \left(\frac{\partial F}{\partial y}\right)^{-1} F(x, y)$ with x fixed near x_0 . The fixed point depends smoothly on x because the derivative of F is continuous. $\square$

This theorem yields local solvability of systems of equations and underpins change-of-variables formulas in multiple integrals, as well as implicit definitions of surfaces and manifolds.

10 Multiple Integrals

Riemann integration extends naturally to higher dimensions and supports integration over regions with curved boundaries.

10.1 Riemann Integration on Rectangles

Let $I = [a_1, b_1] \times \cdots \times [a_n, b_n] \subset \mathbb{R}^n$.

Definition 10.1 (Partitions and Riemann sums). A partition P of I consists of partitions of each coordinate interval. Its mesh is $|P| = \max_i \max_j (x_{i,j} - x_{i,j-1})$. A Riemann sum for $f: I \rightarrow \mathbb{R}$ looks like

$$S(P, f) = \sum f(\xi_{i_1, \dots, i_n}) \prod_{k=1}^n (x_{k, i_k} - x_{k, i_k-1}),$$

with $\xi_{i_1, \dots, i_n}$ chosen from each sub-rectangle.

Upper and lower sums are defined using suprema/infima over each sub-rectangle. Integrability means the sup of the lower sums equals the inf of the upper sums; the common value is $\int_I f$.

10.2 Existence and Volume

Continuous functions on I are integrable. More generally, if f is bounded and continuous except on a set of measure zero, then f is integrable. We integrate over a region E with piecewise smooth boundary by integrating over a containing rectangle and setting $f = 0$ outside E .

The volume of E is $\int_E 1$. Regions with negligible boundary can be approximated by finite unions of rectangles, so their volumes exist.

10.3 Iterated Integrals

Fubini's theorem allows multi-dimensional integrals to be computed as repeated one-dimensional integrals.

Theorem 10.1 (Fubini). *If f is integrable on $[a, b] \times [c, d]$, then*

$$\int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy.$$

The same holds in higher dimensions by iterating the process.

10.4 Change of Variables

Smooth coordinate changes scale integrals by the Jacobian determinant.

Theorem 10.2 (Change of variables). *Let $T: U \rightarrow V$ be a C^1 bijection between open subsets of $\mathbb{R}^n$ with Jacobian determinant J_T . For integrable f on V ,*

$$\int_V f(y) dy = \int_U f(T(x)) |J_T(x)| dx.$$

This identity justifies polar, cylindrical, and spherical coordinates, as well as more general transformations used in evaluating multi-dimensional integrals.